Semester	B.E. Semester VII – Computer Engineering
Subject	DevOps Lab
Subject In-charge	Dr. Snehal Andhare

Student Name	Vicky Pukale
Roll Number	23102A0033
Grade and Subject Teacher's Signature	

Task Number	4
Project Title:	Secure CI/CD Pipeline for a Loan Application Tracking Portal
Task Title	Git and GitHub Repository Initialization: Create a GitHub repository for the Loan Application Tracking Portal. Add README, .gitignore, folder structure, issue templates and branch naming rules. Commit the initial application skeleton using meaningful commit messages.
Deliverable:	**1. Branch Naming Policy & Git Rules**

Branch Type	Naming Convention	Description / Purpose	Target Merge Branch
Production	`main`	Production-ready, stable releases. Tagged with versions (e.g. `v1.0.0`).	Direct commits blocked
Development	`develop`	Integration branch for testing new features before release.	PR required from feature branches
Feature	`feature/<ticket-or-name>`	New capabilities (e.g. `feature/event-entry`, `feature/alerts-view`).	Merged to `develop` via PR
Bugfix	`bugfix/<issue-name>`	Defect fixes during development (e.g. `bugfix/search-filter-null`).	Merged to `develop`
Hotfix	`hotfix/<critical-issue>`	Emergency fixes directly addressing production failures.	Merged to both `main` and `develop`

Commit Message Standard (Conventional Commits)

Commits must follow the format: `<type>(<scope>): <short description>`

- `feat:` A new user-facing feature (e.g., `feat(intake): add loan application submission form`)
- `fix:` A bug fix (e.g., `fix(search): handle case-insensitive application lookup`)
- `test:` Adding or modifying unit or Selenium tests (e.g., `test(e2e): add applicant intake UI test`)
- `ci:` CI/CD pipeline changes (e.g., `ci(jenkins): add Maven test and package stages`)
- `docs:` Documentation updates (e.g., `docs: update setup guide and API specs`)

2. Directory Structure to Create

```
DevOps_Project/
├── .github/
│   └── ISSUE_TEMPLATE/
│       ├── bug_report.md
│       └── feature_request.md
├── src/
│   ├── main/
│   │   ├── java/
│   │   │   └── com/
│   │   │       └── loanportal/
│   │   │           └── LoanApplicationPortalApp.java
│   │   └── resources/
│   │       ├── application.properties
│   │       ├── static/
│   │       │   └── css/
│   │       └── templates/
│   └── test/
│       └── java/
│           └── com/
│               └── loanportal/
│                   └── LoanApplicationPortalAppTests.java
├── .gitignore
├── pom.xml
└── README.md
```

3. Step-by-Step Git Commands to Run on Terminal

Step 1: Verify Maven Build Locally

```
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 4.965 s -- in c
lAppTests
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  37.766 s
[INFO] Finished at: 2026-10-13T14:19:32+05:30
[INFO] ------------------------------------------------------------------------
```

Step 2: Initialize Git and Commit the Baseline

```
D:\Vicky_Project\DevOps\DevOps_Project>git init
Initialized empty Git repository in D:/Vicky_Project/DevOps/DevOps_Project/.git/

D:\Vicky_Project\DevOps\DevOps_Project>git add .
warning: in the working copy of '.github/ISSUE_TEMPLATE/bug_report.md', LF will be replaced by CRLF the next time Git to
uches it
warning: in the working copy of '.github/ISSUE_TEMPLATE/feature_request.md', LF will be replaced by CRLF the next time G
it touches it
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'pom.xml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/main/java/com/loanportal/LoanApplicationPortalApp.java', LF will be replaced by CRL
F the next time Git touches it
warning: in the working copy of 'src/main/resources/application.properties', LF will be replaced by CRLF the next time G
it touches it
warning: in the working copy of 'src/test/java/com/loanportal/LoanApplicationPortalAppTests.java', LF will be replaced b
y CRLF the next time Git touches it

D:\Vicky_Project\DevOps\DevOps_Project>git commit -m "chore: initialize project skeleton with Maven, Spring Boot, and Gi
tHub templates"
[master (root-commit) d9c66ab] chore: initialize project skeleton with Maven, Spring Boot, and GitHub templates
 14 files changed, 237 insertions(+)
 create mode 100644 .github/ISSUE_TEMPLATE/bug_report.md
 create mode 100644 .github/ISSUE_TEMPLATE/feature_request.md
 create mode 100644 .gitignore
 create mode 100644 DOC/DevOps_Task1_Vicky_Pukale_23102A0033.docx
 create mode 100644 DOC/DevOps_Task2_Vicky_Pukale_23102A0033.docx
 create mode 100644 DOC/DevOps_Task3_Vicky_Pukale_23102A0033.docx
 create mode 100644 PDF/DevOps_Task1_Vicky_Pukale_23102A0033.pdf
 create mode 100644 PDF/DevOps_Task2_Vicky_Pukale_23102A0033.pdf
 create mode 100644 PDF/DevOps_Task3_Vicky_Pukale_23102A0033.pdf
 create mode 100644 README.md
 create mode 100644 pom.xml
 create mode 100644 src/main/java/com/loanportal/LoanApplicationPortalApp.java
 create mode 100644 src/main/resources/application.properties
 create mode 100644 src/test/java/com/loanportal/LoanApplicationPortalAppTests.java
```

Step 3: Link to GitHub Repository and Push

```
D:\Vicky_Project\DevOps\DevOps_Project>git branch -M main

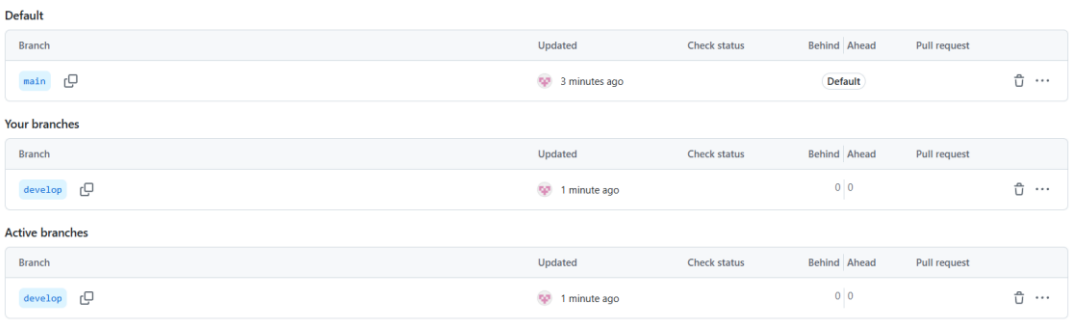
D:\Vicky_Project\DevOps\DevOps_Project>git remote add origin https://github.com/vickypukale/loan-application-tracking-po
rtal.git

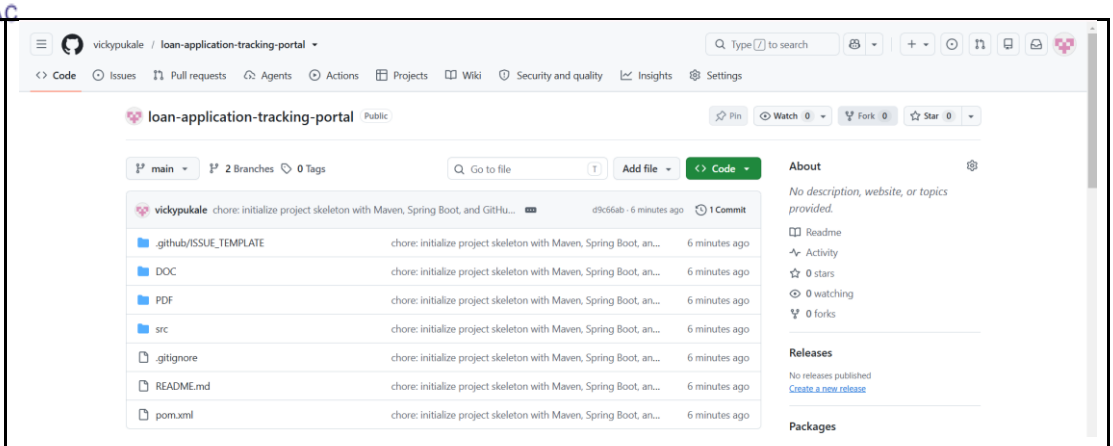
D:\Vicky_Project\DevOps\DevOps_Project>git push -u origin main
Enumerating objects: 30, done.
Counting objects: 100% (30/30), done.
Delta compression using up to 16 threads
Compressing objects: 100% (24/24), done.
Writing objects: 100% (30/30), 1.60 MiB | 1020.00 KiB/s, done.
Total 30 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/vickypukale/loan-application-tracking-portal.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

Step 4: Create and Push the `develop` Branch

```
D:\Vicky_Project\DevOps\DevOps_Project>git checkout -b develop
Switched to a new branch 'develop'

D:\Vicky_Project\DevOps\DevOps_Project>git push -u origin develop
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'develop' on GitHub by visiting:
remote:      https://github.com/vickypukale/loan-application-tracking-portal/pull/new/develop
remote:
To https://github.com/vickypukale/loan-application-tracking-portal.git
 * [new branch]      develop -> develop
branch 'develop' set up to track 'origin/develop'.
```




vickypukale / loan-application-tracking-portal
Type / to search
Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
loan-application-tracking-portal
Public
Pin
Watch 0
Fork 0
Star 0
main
2 Branches
0 Tags
Go to file
Add file
Code
vickypukale chore: initialize project skeleton with Maven, Spring Boot, and GitHu... d9c66ab · 6 minutes ago 1 Commit
.github/ISSUE_TEMPLATE chore: initialize project skeleton with Maven, Spring Boot, an... 6 minutes ago
DOC chore: initialize project skeleton with Maven, Spring Boot, an... 6 minutes ago
PDF chore: initialize project skeleton with Maven, Spring Boot, an... 6 minutes ago
src chore: initialize project skeleton with Maven, Spring Boot, an... 6 minutes ago
.gitignore chore: initialize project skeleton with Maven, Spring Boot, an... 6 minutes ago
README.md chore: initialize project skeleton with Maven, Spring Boot, an... 6 minutes ago
pom.xml chore: initialize project skeleton with Maven, Spring Boot, an... 6 minutes ago
About
No description, website, or topics provided.
Readme
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages